

Задача 5.

8.5.19.

$$\int \frac{dx}{\cos x} = \ln \left| \operatorname{tg} \left(\frac{x}{2} + \frac{\pi}{4} \right) \right| + C$$

8.5.20

$$\int \frac{dx}{1 - \sin x} = \int \frac{dx}{1 - \frac{2 \operatorname{tg} \frac{x}{2}}{1 + \operatorname{tg}^2 \frac{x}{2}}} = \left[t = \operatorname{tg} \frac{x}{2} \quad t' = \frac{1}{2} (1 + \operatorname{tg}^2 \frac{x}{2}) \right] =$$

$$= \int \frac{1}{1 - \frac{2t}{1+t^2}} \cdot \frac{1}{\frac{1}{2}(1+t^2)} dt = \int \frac{1}{\frac{1+t^2-2t}{1+t^2}} \cdot \frac{1}{\frac{1}{2}(1+t^2)} dt =$$

$$= \int \frac{1+t^2}{1+t^2-2t} \cdot \frac{dt}{\frac{1}{2} + \frac{t^2}{2}} = \int \frac{1+t^2}{1+t^2-2t} \cdot \frac{2dt}{1+t^2} =$$

$$= \int \frac{2dt}{1+t^2-2t} = 2 \int \frac{dt}{(1-t)^2} = \left[u = 1-t \quad u' = -1 \quad dt = -du \right] =$$

$$= -2 \int \frac{du}{u^2} = -2 \int u^{-2} du = -2 \cdot \frac{u^{-1}}{-1} + C = \frac{2}{u} + C =$$

$$= \frac{2}{1-t} + C = \frac{2}{1 - \operatorname{tg} \frac{x}{2}} + C$$

8.5.21.

$$\int \frac{dx}{5 + 4 \sin x} = \int \frac{dx}{5 + 4 \left(\frac{2 \operatorname{tg} \frac{x}{2}}{1 + \operatorname{tg}^2 \frac{x}{2}} \right)} = \left[t = \operatorname{tg} \frac{x}{2} \quad t' = \frac{1}{2} (1 + \operatorname{tg}^2 \frac{x}{2}) \quad dx = \frac{1}{t'} dt \right] =$$

$$= \int \frac{dt}{5 + 4 \left(\frac{2t}{1+t^2} \right)} \cdot \frac{1}{\frac{1}{2}(1+t^2)} = \int \frac{1}{5 + \frac{8t}{1+t^2}} \cdot \frac{1}{\frac{1}{2}(1+t^2)} dt =$$

$$\begin{aligned}
 &= \int \frac{\frac{1}{5}(1+t^2)+8t}{1+t^2} \cdot \frac{1}{2(1+t^2)} dt = \int \frac{1+t^2}{5(1+t^2)+8t} \cdot \frac{2dt}{1+t^2} = \\
 &= 2 \int \frac{dt}{5(1+t^2)+8t} = 2 \int \frac{dt}{5+5t^2+8t} = \frac{2}{5} \int \frac{dt}{t^2+\frac{8}{5}t+\frac{16}{25}+\frac{9}{25}} = \\
 &= \frac{2}{5} \int \frac{dt}{(t+\frac{4}{5})^2+\frac{9}{25}} = \left[u = t+\frac{4}{5} \quad u' = 1 \quad dt = du \right] =
 \end{aligned}$$

$$= \frac{2}{5} \int \frac{du}{u^2+\frac{9}{25}} = \frac{2}{5} \cdot \frac{5}{3} \operatorname{arctg} \frac{u}{3} + C =$$

$$\begin{aligned}
 &= \frac{2}{5} \int \frac{dt}{u^2+\frac{9}{25}} = \frac{2}{5} \cdot \frac{5}{3} \operatorname{arctg} \frac{u}{3} + C = \\
 &= \frac{2}{3} \operatorname{arctg} \frac{5(t+\frac{4}{5})}{3} + C = \frac{2}{3} \operatorname{arctg} \frac{5t+4}{3} + C = \\
 &= \frac{2}{3} \operatorname{arctg} \frac{5 \operatorname{tg} \frac{x}{2} + 4}{3} + C
 \end{aligned}$$

5.22

$$\int \frac{2-\sin x}{2+\cos x} dx = \int \frac{2}{2+\cos x} - \frac{\sin x}{2+\cos x} dx = \left[\begin{array}{l} u = 2+\cos x \\ u' = -\sin x \end{array} \right] =$$

$$= 2 \int \frac{1}{2+\frac{1-\operatorname{tg}^2 \frac{x}{2}}{1+\operatorname{tg}^2 \frac{x}{2}}} dx - \int -\frac{1}{u} du = \left[t = \operatorname{tg} \frac{x}{2} \quad t' = \frac{1}{2} (1+\operatorname{tg}^2 \frac{x}{2}) \right] =$$

$$= 2 \int \frac{1}{2+\frac{1-t^2}{1+t^2}} \cdot \frac{1}{2(1+t^2)} dt + \int \ln|u| + C =$$

$$= 2 \int \frac{1}{2+2+t^2+1-t^2} \cdot \frac{2}{1+t^2} dt + \ln|u| + C =$$

$$= 2 \int \frac{1}{2t^2+3} \cdot \frac{2}{1+t^2} dt + \ln|u| + C =$$

$$= 2 \int \frac{1-t^2}{t^2+3} \cdot \frac{2}{1+t^2} dt + \ln|2+\cos x| + C =$$

$$= 4 \int \frac{dt}{t^2+3} + \ln|2+\cos x| + C =$$

$$= 4 \cdot \frac{1}{\sqrt{3}} \operatorname{arctg} \frac{t}{\sqrt{3}} + \ln|2+\cos x| + C =$$

$$= \frac{4}{\sqrt{3}} \operatorname{arctg} \frac{\operatorname{tg} \frac{x}{2}}{\sqrt{3}} + \ln|2+\cos x| + C =$$

$$= \frac{4\sqrt{3}}{3} \operatorname{arctg} \frac{\sqrt{3} \operatorname{tg} \frac{x}{2}}{3} + \ln|2+\cos x| + C$$

8.5.23

$$\int \frac{dx}{2 \sin x - \cos x + 5} = \int \frac{dx}{2 \cdot \frac{2 \operatorname{tg} \frac{x}{2}}{1+\operatorname{tg}^2 \frac{x}{2}} - \frac{1-\operatorname{tg}^2 \frac{x}{2}}{1+\operatorname{tg}^2 \frac{x}{2}} + 5} =$$

$$= \left[t = \operatorname{tg} \frac{x}{2} \quad t' = \frac{1}{2} \left(1 + \operatorname{tg}^2 \frac{x}{2} \right) \right] = \int \frac{1}{2 \cdot \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2} + 5} \cdot \frac{1}{2(1+t^2)} dt =$$

$$= \int \frac{1}{\frac{4t-1+t^2}{1+t^2} + 5} \cdot \frac{2}{1+t^2} dt =$$

$$= \int \frac{1+t^2}{4t-1+t^2+5+5t^2} \cdot \frac{2}{1+t^2} dt = \int \frac{2}{6t^2+4t+4} dt =$$

$$= \int \frac{dt}{3t^2+2t+2} = \frac{1}{3} \int \frac{dt}{t^2+\frac{2}{3}t+\frac{2}{3}} = \frac{1}{3} \int \frac{dt}{t^2+\frac{2}{3}t+\frac{1}{9}+\frac{5}{9}} =$$

$$= \frac{1}{3} \int \frac{dt}{\left(t+\frac{1}{3}\right)^2+\frac{5}{9}} = \left[u = t+\frac{1}{3} \quad u' = 1 \right] = \frac{1}{3} \int \frac{du}{u^2+\frac{5}{9}} =$$

$$= \frac{1}{3} \cdot \frac{3}{\sqrt{5}} \operatorname{arctg} \frac{94}{\sqrt{5}} + C = \frac{\sqrt{5}}{5} \operatorname{arctg} \frac{9\sqrt{5}(t+\frac{1}{3})}{5} + C$$

$$= \frac{\sqrt{5}}{5} \operatorname{arctg} \frac{9\sqrt{5}(\operatorname{tg} \frac{x}{2} + \frac{1}{3})}{5} + C$$

8.5.24

$$\int \frac{1 + \sin x}{(1 + \cos x) \sin x} dx = \int \frac{1 + \sin x}{\sin x + \cos x \sin x} dx =$$

$$= \int \frac{dx}{\sin x + \cos x \sin x} + \int \frac{\sin x}{\sin x + \cos x \sin x} dx =$$

$$= \int \frac{dx}{\sin x + \cos x \sin x} + \int \frac{dx}{1 + \cos x} = \left[t = \operatorname{tg} \frac{x}{2} \quad t' = \frac{1}{2}(1+t^2) \right] =$$

$$= \int \frac{2}{\left(\frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} \cdot \frac{2t}{1+t^2} \right) (1+t^2)} dx + \int \frac{1}{1 + \frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt =$$

$$= \int \frac{1}{\frac{2t}{1+t^2} + \frac{2t(1-t^2)}{(1+t^2)^2}} \cdot \frac{2}{1+t^2} dt + \int \frac{1}{\frac{1+t^2+1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt =$$

$$= \int \frac{1}{\frac{2t+2t-2t^3+2t^3}{(1+t^2)^2}} \cdot \frac{2}{1+t^2} dt + \int \frac{1+t^2}{2} \cdot \frac{2}{1+t^2} dt =$$

$$= \int \frac{1+t^2}{2t} dt + t + C = \int \frac{1}{2t} dt + \int \frac{t^2}{2t} dt + t + C$$

$$= \frac{1}{2} \int \frac{1}{t} dt + \frac{1}{2} \int \frac{t^2}{t} dt + t + C = \frac{1}{2} \ln |t| + \frac{1}{2} \frac{t^2}{2} + t + C =$$

$$= \frac{1}{2} \ln \left| \operatorname{tg} \frac{x}{2} \right| + \frac{\operatorname{tg}^2 \frac{x}{2}}{4} + \operatorname{tg} \frac{x}{2} + C$$

8.5.25

$$\begin{aligned}
 \int \frac{dx}{5\sin^2 X - 3\cos^2 X + 4} &= \left[t = \operatorname{tg} X, dx = \frac{dt}{1+t^2} \right] = \\
 &= \int \frac{dt}{5 \cdot \frac{t^2}{1+t^2} - 3 \cdot \frac{1}{1+t^2} + 4} \cdot \frac{1}{1+t^2} dt = \\
 &= \int \frac{dt}{5t^2 - 3 + 4 + 4t^2} = \int \frac{dt}{9t^2 + 1} = \frac{1}{9} \int \frac{dt}{t^2 + \frac{1}{9}} = \frac{1}{9} \cdot \frac{3}{1} \operatorname{arctg} \frac{3t}{1} + C = \\
 &= \frac{1}{3} \operatorname{arctg} 3 \operatorname{tg} X + C
 \end{aligned}$$

8.5.26

$$\begin{aligned}
 \int \frac{dx}{4\sin^2 X + 9\cos^2 X} &= \left[t = \operatorname{tg} X, dx = \frac{1}{1+t^2} dt \right] = \\
 &= \int \frac{dt}{4 \cdot \frac{t^2}{1+t^2} + 9 \cdot \frac{1}{1+t^2}} \cdot \frac{1}{1+t^2} = \int \frac{dt}{\frac{4t^2 + 9}{1+t^2}} \cdot \frac{1}{1+t^2} = \\
 &= \int \frac{1+t^2}{4t^2 + 9} \cdot \frac{1}{1+t^2} dt = \frac{1}{4} \int \frac{dt}{t^2 + \frac{9}{4}} = \frac{1}{4} \cdot \frac{2}{3} \operatorname{arctg} \frac{2t}{3} + C = \\
 &= \frac{1}{6} \operatorname{arctg} \frac{2}{3} \operatorname{tg} X + C
 \end{aligned}$$

8.5.27

$$\begin{aligned}
 \int \frac{dx}{1+3\cos^2 X} &= \left[t = \operatorname{tg} X, dx = \frac{1}{1+t^2} dt \right] = \int \frac{dx}{1+3 \cdot \frac{1}{1+t^2}} \cdot \frac{1}{1+t^2} = \\
 &= \int \frac{dt}{\frac{1+t^2+3}{1+t^2}} \cdot \frac{1}{1+t^2} = \int \frac{1+t^2}{t^2+4} \cdot \frac{1}{1+t^2} dt = \\
 &= \int \frac{dt}{t^2+4} = \frac{1}{2} \operatorname{arctg} \frac{t}{2} + C = \frac{1}{2} \operatorname{arctg} \frac{\operatorname{tg} X}{2} + C
 \end{aligned}$$

8.5.28

$$\int \frac{dx}{\sin^4 x} = \left[t = \operatorname{tg} x, dx = \frac{1}{1+t^2} dt \right] = \int \frac{dt}{\left(\frac{1}{1+t^2} \right)^2} \cdot \frac{1}{t^2+1} =$$

$$\int \frac{(1+t^2)^2}{1} \cdot \frac{1}{t^2+1} dt = \int 1+t^2 dt =$$

$$= \int 1 dt + \int t^2 dt = t + \frac{t^3}{3} + C = \operatorname{tg} x + \frac{1}{3} \operatorname{tg}^3 x + C$$

8.5.29

$$\int \sin^5 x dx = \int (1 - \cos^2 x)^2 \sin x dx = \left[t = \cos x, dx = -\frac{1}{\sin x} dt \right] =$$

$$= - \int (1-t^2)^2 dt = - \int 1 - 2t^2 + t^4 dt = - \left(\int 1 dt - \int 2t^2 dt + \int t^4 dt \right) =$$

$$= - \left(\frac{t^5}{5} - 2 \frac{t^3}{3} + t \right) + C = -\frac{1}{5} \cos^5 x + \frac{2}{3} \cos^3 x + \cos x + C$$

8.5.30

$$\int \sin^4 x \cdot \cos^5 x dx = \left[t = \sin x, dx = \frac{1}{\sqrt{1-t^2}} dt = \frac{1}{\cos x} dt \right] =$$

$$= \int t^4 \cdot (1-t^2)^2 dt = \int t^4 (1-2t^2+t^4) dt =$$

$$= \int t^4 - 2t^6 + t^8 dt = \int t^4 dt - 2 \int t^6 dt + \int t^8 dt =$$

$$= \frac{1}{5} t^5 - \frac{2}{7} t^7 + \frac{1}{9} t^9 + C = \frac{1}{5} \sin^5 x - \frac{2}{7} \sin^7 x + \frac{1}{9} \sin^9 x + C$$

8.5.31

$$\int \frac{\sin 2x dx}{\cos^7 x} = \int \frac{2 \sin x \cos x dx}{\cos^7 x} = 2 \int \frac{\sin x}{\cos^6 x} dx =$$

$$= 2 \left[t = \cos x, dx = -\frac{1}{\sin x} dt \right] = -2 \int \frac{\sin x}{t^6} \cdot \frac{1}{\sin x} dt =$$

$$= -2 \int t^{-6} dt = -2 \frac{t^{-5}}{-5} + C = \frac{2}{5} \cos^5 x + C$$

8.5.32.

$$\int \frac{\sin^4 x \, dx}{\cos x} = \text{[scribbled out]}$$

$$= \int \frac{\sin^4 x \cos x}{\cos^2 x} dx = \int \frac{\sin^4 x \cos x}{1 - \sin^2 x} dx =$$

$$= [t = \sin x, \, dx = \frac{1}{\cos x} dt] = \int \frac{t^4}{1-t^2} dt = - \int \frac{t^4}{t^2-1} dt =$$

$$= - \int \frac{1}{t^2-1} + t^2 + 1 \, dt = \int \frac{1}{t^2-1} dt + \int t^2 dt + \int 1 dt =$$

$$= \frac{1}{2} \ln \left| \frac{t-1}{t+1} \right| + \frac{1}{3} t^3 + t + C =$$

$$= \frac{1}{2} \ln \left| \frac{\sin x - 1}{\sin x + 1} \right| + \frac{1}{3} \sin^3 x + \sin x + C$$

8.5.33

$$\int \sin^6 x \, dx = \int \left(\frac{1 - \cos 2x}{2} \right)^3 dx = \int \frac{(1 - \cos 2x)^3}{8} dx =$$

$$= \frac{1}{8} \left(\int 1 - 3\cos 2x + 3\cos^2 2x - \cos^3 2x \, dx \right) =$$

$$= \frac{1}{8} x - \frac{3}{8} \sin 2x + \frac{3}{8} \frac{\sin^3 2x}{3} - \frac{1}{8} \frac{\sin^4 2x}{4} + C =$$

$$= \frac{1}{8} x - \frac{3}{8} \sin 2x + \frac{1}{8} \sin^3 2x - \frac{1}{32} \sin^4 2x + C$$

8.5.34

$$\int \sin^2 x \cos^4 x \, dx = \int \frac{1 + \cos 2x}{2} \cdot \frac{(1 - \cos 2x)^2}{4} dx =$$

$$= \int \frac{(1 + \cos 2x)(1 - \cos 2x)^2}{8} dx = [u = 2x \quad u' = 2 \quad dx = \frac{1}{2} du] =$$

$$= \frac{1}{16} \int (1 + \cos u)(1 - \cos u)^2 du = \frac{1}{16} \int \cos^3 u - \cos^2 u -$$

$$- \cos u + 1 \, du = \frac{1}{16} \frac{\cos^4 u}{4} - \frac{1}{16} \frac{\cos^3 u}{3} - \frac{1}{16} \frac{\cos^2 u}{2} + \frac{u}{16} + C =$$

$$= \frac{\cos^4 2x}{64} - \frac{\cos^3 2x}{48} - \frac{\cos^2 2x}{32} + \frac{x}{8} + C$$

8.5.35

$$\int \sin^4 x \cos^4 x \, dx = \int \left(\frac{1+\cos 2x}{2} \right)^2 \left(\frac{1-\cos 2x}{2} \right)^2 dx =$$

$$= \frac{1}{16} \int (1+\cos 2x)^2 (1-\cos 2x)^2 dx =$$

$$= \frac{1}{16} \int (1+2\cos 2x+\cos^2 2x)(1-2\cos 2x+\cos^2 2x) dx =$$

$$= \frac{1}{16} \int 1-2\cos 2x+\cos^2 2x+2\cos 2x-4\cos^2 2x+2\cos^3 2x+\cos^2 2x-2\cos^3 2x+\cos^4 2x \, dx =$$

$$= \frac{1}{16} \int \cos^4 2x - 2\cos^2 2x + 1 \, dx =$$

$$= \frac{1}{16} \frac{\cos^5 2x}{5} - \frac{1}{8} \frac{\cos^3 2x}{3} + \frac{1}{16} x + C = \frac{\cos^5 2x}{80} -$$

$$- \frac{\cos^3 2x}{24} + \frac{1}{16} x + C$$

8.5.36

$$\int \sin x \cdot \sin 3x \, dx = \int \frac{1}{2} (\cos(x-3x) - \cos(x+3x)) \, dx =$$

$$= \frac{1}{2} \int \cos 2x - \cos 4x \, dx = \left[\begin{array}{l} u=2x \quad dx=\frac{1}{2} du \\ t=4x \quad dx=\frac{1}{4} dt \end{array} \right] =$$

$$= \frac{1}{2} \int \frac{\cos u}{2} du - \frac{1}{2} \int \frac{\cos t}{4} dt =$$

$$= \frac{1}{4} \sin 2x - \frac{1}{8} \sin 4x + C$$

8.5.37

$$\begin{aligned}
 \int \sin \frac{x}{12} \cos \frac{x}{3} dx &= \int \frac{1}{2} \left(\sin \left(\frac{x}{12} - \frac{4x}{12} \right) + \sin \left(\frac{x}{12} + \frac{4x}{12} \right) \right) dx = \\
 &= \frac{1}{2} \int \sin \left(-\frac{x}{4} \right) + \sin \left(\frac{5x}{4} \right) dx = -\frac{1}{2} \int \sin \frac{x}{4} dx + \frac{1}{2} \int \sin \frac{5x}{4} dx = \\
 &= \left[u = \frac{x}{4} \quad u' = \frac{1}{4} \quad dx = 4 du \right] = \\
 &= -\frac{1}{2} \int \sin u \cdot 4 du + \frac{1}{2} \int \sin 5u \cdot 4 du = \\
 &= -2 \int \sin u du + 2 \int \sin 5u du = 2 \cos u + 2 \cos 5u + C = \\
 &= 4 \cos \frac{x}{4} + C
 \end{aligned}$$

8.5.38

$$\begin{aligned}
 \int \cos x \cdot \cos 3x dx &= \int \frac{1}{2} (\cos 2x + \cos 4x) dx = \\
 &= \frac{1}{2} \int \cos 2x dx + \frac{1}{2} \int \cos 4x dx = \left[u = 2x, \quad dx = \frac{1}{2} du \right] - \\
 &\quad t = 4x, \quad dx = \frac{1}{4} du
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2} \int \cos u \cdot \frac{1}{2} du + \frac{1}{2} \int \cos t \cdot \frac{1}{4} du = \\
 &= \frac{1}{4} \sin u + \frac{1}{8} \sin t = \frac{1}{4} \sin 2x + \frac{1}{8} \sin 4x + C
 \end{aligned}$$

8.5.39

$$\begin{aligned}
 \int \cos x \cdot \cos 3x \cdot \cos 5x dx &= \int \frac{1}{2} (\cos 2x + \cos 4x) \cos 5x dx = \\
 &= \frac{1}{2} \int \cos 2x \cos 5x + \cos 4x \cos 5x dx = \\
 &= \frac{1}{2} \int \frac{1}{2} (\cos 3x + \cos 7x) + \frac{1}{2} (\cos x + \cos 9x) dx = \\
 &= \frac{1}{4} \int \cos 3x + \cos 7x + \cos x + \cos 9x dx =
 \end{aligned}$$

$$= \frac{1}{4} \left(\frac{\sin 3x}{3} + \frac{\sin 7x}{7} + \sin x + \frac{\sin 9x}{9} \right) + C =$$

$$= \frac{1}{4} \frac{\sin 3x}{12} + \frac{\sin 7x}{28} + \frac{\sin x}{4} + \frac{\sin 9x}{36} + C$$

8.5.40

$$\int \operatorname{ctg}^6 x \, dx = \int \operatorname{ctg}^4 x \left(\frac{1}{\sin^2 x} - 1 \right) dx =$$

$$= \int \frac{\operatorname{ctg}^4 x}{\sin^2 x} - \operatorname{ctg}^4 x \, dx = \int \frac{\operatorname{ctg}^4 x}{\sin^2 x} \, dx - \int \operatorname{ctg}^4 x \, dx =$$

$$= [u = \operatorname{ctg} x \quad dx = -\sin^2 x] =$$

$$= \int -u^4 \, du - \int \operatorname{ctg}^4 x \, dx = -\frac{u^5}{5} + C - \int \operatorname{ctg}^4 x \, dx =$$

$$= -\frac{u^5}{5} + C - \int \operatorname{ctg}^2 x \left(\frac{1}{\sin^2 x} - 1 \right) dx =$$

$$= -\frac{\operatorname{ctg}^5 x}{5} + C - \int \frac{\operatorname{ctg}^2 x}{\sin^2 x} - \operatorname{ctg}^2 x \, dx = [u = \operatorname{ctg} x \, dx = -\sin^2 x] =$$

$$= -\frac{\operatorname{ctg}^5 x}{5} + C + \int u^2 - \int \operatorname{ctg}^2 x \, dx =$$

$$= -\frac{\operatorname{ctg}^5 x}{5} + \frac{\operatorname{ctg}^3 x}{3} - \int \operatorname{ctg}^2 x \, dx =$$

$$= -\frac{\operatorname{ctg}^5 x}{5} + \frac{\operatorname{ctg}^3 x}{3} - \int \frac{1}{\sin^2 x} - 1 \, dx =$$

$$= -\frac{\operatorname{ctg}^5 x}{5} + \frac{\operatorname{ctg}^3 x}{3} - \operatorname{ctg} x - x + C$$

8.5.41

$$\int \operatorname{tg}^4 \frac{x}{2} dx = \left[u = \frac{x}{2} \quad u' = \frac{1}{2} \quad dx = 2du \right] =$$

$$= 2 \int \operatorname{tg}^4 u \, du = 2 \int \left(\frac{1}{\cos^2 u} - 1 \right) \operatorname{tg}^2 u \, du =$$

$$= 2 \int \frac{\operatorname{tg}^2 u}{\cos^2 u} - \operatorname{tg}^2 u \, du = 2 \int \frac{\operatorname{tg}^2 u}{\cos^2 u} \, du - 2 \int \operatorname{tg}^2 u \, du =$$

$$= \left[t = \operatorname{tg} u, \quad du = \frac{1}{\cos^2 u} \, du \right] =$$

$$= 2 \int t^2 \, du - 2 \int \frac{1}{\cos^2 u} - 1 \, du =$$

$$= 2 \frac{t^3}{3} - 2 \operatorname{tg} u + 2u + C = \frac{2}{3} \frac{\operatorname{tg}^3 u}{\cos^3 u} - 2 \operatorname{tg} \frac{x}{2} +$$

$$+ 2 \cdot \frac{x}{2} + C = \frac{2}{3} \operatorname{tg}^3 \frac{x}{2} - 2 \operatorname{tg} \frac{x}{2} + x + C$$

8.5.42

$$\int \operatorname{tg}^7 x \, dx = \int \frac{\sin^7 x}{\cos^7 x} \, dx = \int \frac{(1 - \cos^2 x)^3 \sin x}{\cos^7 x} \, dx =$$

$$= \left[u = \cos x \quad dx = -\frac{1}{\sin x} \, du \right] = - \int \frac{(1 - u^2)^3}{u^7} \, du =$$

$$= - \int \frac{-u^6 + 3u^4 - 3u^2 + 1}{u^7} \, du = - \int -\frac{1}{u} + \frac{3}{u^3} - \frac{3}{u^5} + \frac{1}{u^7} \, du$$

$$= \ln|u| - 3 \frac{u^{-2}}{-2} + 3 \frac{u^{-4}}{-4} - \frac{u^{-6}}{-6} + C =$$

$$= \ln|\cos x| + \frac{3}{2 \cos^2 x} - \frac{3}{4 \cos^4 x} + \frac{1}{6 \cos^6 x} + C$$